5.2.1 Show when $\Gamma_1 = \Gamma_2 = \Gamma_1(N)$ and $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$,

$\Gamma_3 = \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2 = \Gamma_1^0(N, p)$.

Pf: $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2 = \left\{ X \equiv \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{p} \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} (N) \right\} \cap \Gamma_1(N)$

$= \left\{ X \equiv \begin{pmatrix} 1 & b \\ 0 & \frac{1}{p} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} (N) \right\} \cap \left\{ X \equiv \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} (N) \right\}$

$= \left\{ X \equiv \begin{pmatrix} 1 & pb \\ 0 & 1 \end{pmatrix} (N) \right\} \cap \left\{ X \equiv \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} (N) \right\}$

$= \Gamma_p(N) \cap \Gamma^0(p) = \Gamma_0'(N, p)$.

5.2.2 When $\gamma_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $p \mid a$, then $\gamma_2 \gamma_{2,\infty}^{-1} \in \Gamma_3$.

Pf: $\gamma_{2,\infty}$ is defined as $\gamma_{2,\infty} = \begin{pmatrix} mp & n \\ N & 1 \end{pmatrix}$ $mp - nN = 1$.

Remark: $\gamma_2 \in \Gamma_2 = \Gamma_1(N) \Rightarrow a = nN + 1$. $p \mid a \Rightarrow mp = a$.

Now, ~~$\gamma_{2,\infty} =$~~ $mp \equiv 1$ ~~(N)~~ (nN) $\Rightarrow mp \equiv 1\ (N)$ ~~thus~~

thus $\gamma_{2,\infty} \in \Gamma_2$. hence $\gamma_2 \gamma_{2,\infty}^{-1} \in \Gamma_2$

Compute ~~upper right entry $b' =$ of~~ $\gamma_{2,\infty}^{-1}$:

$\begin{pmatrix} mp & n \\ N & 1 \end{pmatrix} \Leftrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 \\ N & 1 \end{pmatrix} \Leftrightarrow \begin{pmatrix} 1 & -n \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Leftrightarrow \begin{pmatrix} 1 & -n \\ -N & 1+nN \end{pmatrix}$

Thus upper right entry of $\gamma_2 \gamma_{2,\infty}^{-1}$ is $b' = -an + b(1+nN) = -an + bmp$.

Thus $p \mid b'$. $\gamma_2 \gamma_{2,\infty}^{-1} \in \Gamma^0(p)$. Thus it is in Γ_3.

5.2.3 Show $\gamma_{2,j}$ $j = 0, \cdots, p-1$ ~~(∞)~~ $p \nmid a$ represents different cosets.
$0, \cdots, p-1, \infty$ $p \mid a$

Pf: We want to show $\Gamma_3 \gamma_{2,j} \cap \Gamma_3 \gamma_{2,j'} = \phi$ $j \not\equiv j'(p)$

That is $\gamma_3 \gamma_{2,j} = \gamma_{2,j'} \Rightarrow j \equiv j'(p)$ $\gamma_3 \in \Gamma_3$.

Let $\gamma_3 \equiv \begin{pmatrix} 1 & tp \\ \cancel{tN} 0 & 1 \end{pmatrix} (N)$ then $\begin{pmatrix} 1 & tp \\ \cancel{tN} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix} \equiv \begin{pmatrix} 1 & j+tp \\ \cancel{tN} 0 & \cancel{tN+1} \end{pmatrix} \equiv \begin{pmatrix} 1 & j' \\ 0 & 1 \end{pmatrix} (N)$
$\Rightarrow j \equiv j'(p)$.

$\begin{pmatrix} 1 & tp \\ 0 & 1 \end{pmatrix} \begin{pmatrix} mp & n \\ N & 1 \end{pmatrix} \equiv \begin{pmatrix} mp & n+tp \\ N & 1 \end{pmatrix} (N)$. $n \equiv n\ (p)$.

5.24 Show $\Gamma_1 = \Gamma_2 = \Gamma_0(N)$, $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$ gives the same orbit representatives $\{\beta_j\}$ for $\Gamma_1 \backslash \Gamma_1 \alpha \Gamma_2$.

Pf: Given $\gamma_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_2$

representatives of $\Gamma_1 \backslash \Gamma_1 \alpha \Gamma_2$ are reps of $\Gamma_3 \backslash \Gamma_2$ left multiplied by α.

$\Gamma_3 = \alpha^{-1} \Gamma_1 \alpha \cap \Gamma_2 = \cancel{\Gamma_0(N)}.\ \Gamma_0^p(N, p)$.

since $\begin{pmatrix} 1 & \\ & \frac{1}{p} \end{pmatrix} \begin{pmatrix} a & b \\ kN & d \end{pmatrix} \begin{pmatrix} 1 & \\ & p \end{pmatrix} = \begin{pmatrix} a & pb \\ \frac{1}{p} kN & d \end{pmatrix}$.

$\gamma_2 \in \Gamma_3 \gamma_{2,j}$ if $\gamma_2 \gamma_{2,j}^{-1} \in \Gamma_3 = \Gamma_2 \cap \Gamma^0(p)$.

$\gamma_{2,j} = \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix}$ $j = 0, \cdots, p-1$ and $\gamma_{2,\infty} = \begin{pmatrix} mp & n \\ N & 1 \end{pmatrix}$ defined as before does.

We also need $\gamma_2 \gamma_{2,j}^{-1}$ has upper right entry $\equiv 0 \ (p)$.

as they do have for $\Gamma_1 = \Gamma_2 = \Gamma_1(N)$.

Thus $\gamma_2 \gamma_{2,j}^{-1} \in \Gamma_3$. They give representatives

$\beta_j = \alpha \gamma_{2,j} = \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix}$ and $\beta_\infty = \alpha \gamma_{2,\infty} = \begin{pmatrix} m & n \\ N & p \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$.

$f \in \mathcal{M}_k(\Gamma_0(N)) \Rightarrow f\left[\begin{pmatrix} m & n \\ N & p \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}\right]_k = f\left[\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}\right]_k$.

Prove Prop 5.2.3 (generalized divisor sum).

5.2.5 (a) Prove $\sigma_{k-1}^{\psi,\varphi}$ is multiplicative if input is coprime

Pf: $\sigma_{k-1}^{\psi,\varphi}(n) = \sum_{\substack{i|n \\ i>0}} \psi(n/i)\,\varphi(i)\, i^{k-1}$ $\qquad \gcd(m,n)=1$

$\sigma_{k-1}^{\psi,\varphi}(m) = \sum_{\substack{j|m \\ j>0}} \psi(m/j)\,\varphi(j)\, j^{k-1}$.

Thus $\sigma_{k-1}^{\psi,\varphi}(n)\,\sigma_{k-1}^{\psi,\varphi}(m) = \sum_{\substack{ij|nm \\ ij>0}} \psi(n/i)\,\psi(m/j)\,\varphi(i)\varphi(j)(ij)^{k-1}$

uses $\gcd(m,n)=1$

characters are multiplicative

$$\sum_{\ell|nm} \psi\left(\frac{nm}{\ell}\right)\varphi(\ell)\,\ell^{k-1} = \sigma_{k-1}^{\psi,\varphi}(nm).$$

(b) i) Let p prime, $n \geq 1$. Write $n = n' p^e$ $\quad p \nmid n'$, $e \geq 0$.

Show $\sigma_{k-1}^{\psi,\varphi}(np) = \psi(p)\,\sigma_{k-1}^{\psi,\varphi}(n) + \varphi(p^{e+1})(p^{e+1})^{k-1}\,\sigma_{k-1}^{\psi,\varphi}(n')$

Pf: $\sigma_{k-1}^{\psi,\varphi}(np) = \sum_{\substack{m|np \\ m>0}} \psi(np/m)\,\varphi(m)\,m^{k-1} = \sum_{\substack{m|n'p^{e+1} \\ m>0}} \psi(n'p^{e+1}/m)\,\varphi(m)\,m^{k-1}$.

$= \sum_{i=1}^{e+1} \psi(n'p^{e+1}/p^i)\,\varphi(p^i)\,(p^i)^{k-1} + \sum_{\substack{m|n' \\ m>0}} \psi(n'p^{e+1}/m)\,\varphi(m)\,m^{k-1}$.

RHS $= \psi(p) \sum_{\substack{m|n'p^e \\ m>0}} \psi(n'p^e/m)\,\varphi(m)\,m^{k+1} + \varphi(p^{e+1})(p^{e+1})^{k-1} \sum_{\substack{\ell|n' \\ \ell>0}} \psi(n'/\ell)\,\varphi(\ell)\,\ell^{k+1}$

$= \sum_{\substack{m|n'p^e \\ m>0}} \psi(np/m)\,\varphi(m)\,m^{k+1} + \sum_{\substack{\ell|n' \\ \ell>0}} \psi(n'/\ell)\,\varphi(\ell p^{e+1})(\ell p^{e+1})^{k+1}$

$= \sum_{\substack{m|n'p^e \\ m>0}} \psi(np/m)\,\varphi(m)\,m^{k+1} + \sum_{\substack{\ell p^{e+1}|np^{e+1} \\ \ell>0}} \psi(np/\ell p^{e+1})\,\varphi(\ell p^{e+1})(\ell p^{e+1})^{k+1} = \sigma_{k-1}^{\psi,\varphi}(np)$.

ii) Also use (a) to show when $e \geq 1$, $p \nmid N$,

$$\chi(p)\, p^{k-1} \sigma_{k-1}^{\psi,\varphi}(n/p) = \varphi(p)\, p^{k-1} \sigma_{k-1}^{\psi,\varphi}(n) - \varphi(p^{e+1})(p^{e+1})^{k-1} \sigma_{k-1}^{\psi,\varphi}(n').$$

Pf: By i), $\sigma_{k-1}^{\psi,\varphi}(n) = \psi(p)\, \sigma_{k-1}^{\psi,\varphi}(n/p) + \varphi(p^e)(p^e)^{k-1} \sigma_{k-1}^{\psi,\varphi}(n')$.

Thus $\chi(p)\, p^{k-1} \sigma_{k-1}^{\psi,\varphi}(n/p) = \varphi(p)\, p^{k-1} \sigma_{k-1}^{\psi,\varphi}(n) - \varphi(p^{e+1})(p^{e+1})^{k-1} \sigma_{k-1}^{\psi,\varphi}(n')$.

$\psi(p)\varphi(p) = \chi(p)$ (Where did we used (a)?).

iii) Use the two to show $\forall\, e \geq 0$ $p \nmid N$,

$$\sigma_{k-1}^{\psi,\varphi}(np) + \chi(p)\, p^{k-1} \sigma_{k-1}^{\psi,\varphi}(n/p) = \left(\psi(p) + \varphi(p)\, p^{k-1}\right) \sigma_{k-1}^{\psi,\varphi}(n)$$

Pf: LHS $= \psi(p)\, \sigma_{k-1}^{\psi,\varphi}(n) + \varphi(p^{e+1})(p^{e+1})^{k-1} \sigma_{k-1}^{\psi,\varphi}(n')$

$$+ \varphi(p)\, p^{k-1} \sigma_{k-1}^{\psi,\varphi}(n) - \varphi(p^{e+1})(p^{e+1})^{k-1} \sigma_{k-1}^{\psi,\varphi}(n')$$

$$= \left(\psi(p) + \varphi(p)\, p^{k-1}\right) \sigma_{k-1}^{\psi,\varphi}(n).$$

iv) Show this formula also holds $\forall\, e \geq 0$, $p \mid N$.

5.2.7 (a) i) Show $\forall \tau \in \mathcal{H}$ $\forall \beta_j = \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix}$ $0 \le j < p$

$$C = \Lambda_{\beta_j(\tau)} = \langle (\tau+j)/p \rangle + \Lambda_\tau .$$

Pf: $\Lambda_{\beta_j(\tau)} = \beta_j(\tau)\mathbb{Z} \oplus \mathbb{Z} = (\tau+j)/p\, \mathbb{Z} \oplus \mathbb{Z}$.

$\Lambda_\tau = \tau\mathbb{Z} \oplus \mathbb{Z}$.

$\langle (\tau+j)/p \rangle + \Lambda_\tau = \{ (\tau+j)/p + m\tau + n : m,n \in \mathbb{Z} \} = (\tau+j)/p\, \mathbb{Z} \oplus \tau\mathbb{Z} \oplus \mathbb{Z}$.

$= \{ \cancel{(\tau+j)/p + n} \; \cancel{m\tau} + pm\tau/p + j + pn/p : m,n \in \mathbb{Z} \}$

$p\Lambda_{\beta_j(\tau)} = (\tau+j)\mathbb{Z} \oplus p\mathbb{Z} = \cancel{\tau\mathbb{Z} \oplus p\mathbb{Z} \oplus j\mathbb{Z}}$.

$\{ a\tau + bp + aj : a,b \in \mathbb{Z} \}$

We could see $\Lambda_{\beta_j(\tau)} \subset \langle (\tau+j)/p \rangle + \Lambda_\tau$, while

$\forall a,b,c$ $a(\tau+j)/p + b\tau + c = (a+bp)(\tau+j)/p + (c-bj) \in \Lambda_{\beta_j(\tau)}$.

Thus $\Lambda_{\beta_j(\tau)} = \langle (\tau+j)/p \rangle + \Lambda_\tau$.

ii) Show C satisfies (5.7) $C \simeq \mathbb{Z}/p\mathbb{Z}$ in E_τ.

$C \cap (\langle 1/N \rangle + \Lambda_\tau) = \{0\}$ in E_τ.

Pf: $pC = (\tau+j)\mathbb{Z} \oplus p\mathbb{Z} \overset{\gcd(j,p)=1}{=} \tau\mathbb{Z} \oplus \mathbb{Z} = \Lambda_\tau$.

Thus $pC \sim 0$ in E_τ.

$C \cap (\langle 1/N \rangle + \Lambda_\tau) = (\langle (\tau+j)/p \rangle + \Lambda_\tau) \cap (\langle 1/N \rangle + \Lambda_\tau)$

$= (\langle (\tau+j)/p \rangle \cap \langle 1/N \rangle) + \Lambda_\tau$

$m(\tau+j)/p = n/N \Rightarrow m = n = 0$.

Thus $C \cap (\langle 1/N \rangle + \Lambda_\tau) = \{0\} + \Lambda_\tau \sim \{0\}$ in E_τ.

(5.2.7 (a)) iii) Show $\forall \tau \in \mathcal{H}$ $p \nmid N$ $\beta_\infty = \begin{pmatrix} m & n \\ N & p \end{pmatrix}\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$

$$C = C_\infty = (N\tau+1)\Lambda_{\beta_\infty(\tau)} = \langle 1/p \rangle + \Lambda_\tau.$$

Pf: $\langle 1/p \rangle + \Lambda_\tau = 1/p\,\mathbb{Z} \oplus \tau\mathbb{Z} \oplus \mathbb{Z} = 1/p\,\mathbb{Z} \oplus \tau\mathbb{Z}.$

$$(N\tau+1)\Lambda_{\beta_\infty(\tau)} = (N\tau+1)\beta_\infty(\tau)\mathbb{Z} \oplus (N\tau+1)\mathbb{Z}$$

$$= (N\tau+1)\frac{m(p\tau)+n}{N(p\tau)+p}\mathbb{Z} \oplus (N\tau+1)\mathbb{Z} = \frac{(mp\tau+n)}{p}\mathbb{Z} \oplus (N\tau+1)\mathbb{Z}$$

$$= m\tau\mathbb{Z} \oplus n/p\,\mathbb{Z} \oplus N\tau\mathbb{Z} \oplus \mathbb{Z}.$$

$mp - nN = 1 \Rightarrow \gcd(m, N) = 1$, $\gcd(n, p) = 1$.

Thus $(N\tau+1)\Lambda_{\beta_\infty(\tau)} = \tau\mathbb{Z} \oplus \frac{kn+lp}{p}\mathbb{Z} = \tau\mathbb{Z} \oplus 1/p\,\mathbb{Z}$

Thus $C_\infty = \langle 1/p \rangle + \Lambda_\tau = (N\tau+1)\Lambda_{\beta_\infty(\tau)}$.